ratchet and derive the new key as the encryption of an all-zero message using that reserved IV, as shown in listing 12.17 using AES in Counter mode. In this example, a 128-bit IV of all 1-bits is reserved for the ratchet operation as it is highly unlikely that this value would be generated by either a counter or a randomly generated IV.

> **WARNING** You should ensure that the special IV used for the ratchet is never used to encrypt a message.

Listing 12.17 Ratcheting with AES-CTR

```
private static byte[] ratchet(byte[] oldKey) throws Exception
{
    var cipher = Cipher.getInstance("AES/CTR/NoPadding");
    var iv = new byte[16];    #A
    Arrays.fill(iv, (byte) 0xFF);    #A
    cipher.init(Cipher.ENCRYPT_MODE,
            new SecretKeySpec(oldKey, "AES"),     #B
            new IvParameterSpec(iv));    #B
    return cipher.doFinal(new byte[32]);   #C
}
```

#A Reserve a fixed IV that is only used for ratcheting
#B Initialize the cipher using the old key and the fixed ratchet IV
#C Encrypt 32 zero bytes and use the output as the new key

After performing a ratchet, you should ensure the old key is scrubbed from memory so that it can't be recovered, as shown in the following example:

```
var newKey = ratchet(key);
Arrays.fill(key, (byte) 0);    #A
```